

Decision Trees by Hand

Learning goals

After this exercise, you should be able to

- explain what a regression tree predicts inside a leaf,
- explain why the mean is the optimal constant prediction when squared error is used,
- interpret residuals, mean squared error (MSE), and sum of squared errors (SSE),
- describe how alternative loss functions would change the value predicted by a leaf,
- determine all relevant candidate split points for a one-dimensional feature,
- reproduce the main steps that a regression-tree training algorithm performs when choosing a split,
- construct a shallow regression tree by hand,
- explain why a regression tree produces a piecewise-constant prediction,
- explain qualitatively how increasing tree depth affects approximation quality and overfitting,
- extend the same reasoning to a dataset with more than one input feature.

The calculations in this sheet are designed to be done by hand. No programming is required.

Part I - Understanding a single leaf

A regression tree partitions the input space into regions. Each final region is called a leaf. All observations that arrive in the same leaf receive the same prediction.

Consider a leaf that contains four target values

observation	target y_i
1	2
2	3
3	4
4	7

Exercise 1.1 - One constant prediction

Assume that the leaf predicts one constant value c for all four observations, i.e. we construct a tree with only a single leaf.

1. Compute the mean of the four target values. This will be the value of the leaf output.
2. Use this mean as the prediction c .
3. Compute the residual

$$r_i = y_i - c$$

for every observation.

4. Compute the sum of squared errors

$$\text{SSE}(c) = \sum_i (y_i - c)^2$$

5. Compute the mean squared error

$$\text{MSE}(c) = \frac{1}{N} \sum_i (y_i - c)^2$$

What information does the MSE give you about the predictions in this leaf?

Exercise 1.2 - Is the mean special?

Consider the squared-error loss as a function of an arbitrary constant c :

$$L(c) = \sum_{i=1}^N (y_i - c)^2$$

1. Differentiate $L(c)$ with respect to c .
2. Set the derivative equal to zero to obtain a local extremum.
3. Show that the minimizer is

$$c = \frac{1}{N} \sum_i y_i$$

Explain in words why a regression-tree leaf predicts the mean target value when a squared error is used.

Exercise 1.3 - What does MSE actually measure?

Discuss the following questions (short answer only).

1. Is the MSE measured in the same units as the target y ?
2. Why do large residuals contribute particularly strongly to the MSE?
3. What would an MSE of zero mean?
4. Can two very different sets of residuals have the same MSE?
5. The root mean squared error is defined as

$$\text{RMSE} = \sqrt{\text{MSE}}$$

What advantage does RMSE have when interpreting the size of a typical prediction error?

Exercise 1.4 - Could we use another criterion?

Squared error is not the only possible loss.

Consider instead the mean absolute error

$$\text{MAE}(c) = \frac{1}{N} \sum_i |y_i - c|$$

You do not need to perform a full minimization.

1. Which quantity replaces the mean as the natural constant prediction when absolute error is minimized?
 2. Which loss, MSE or MAE, reacts more strongly to a very large outlier?
 3. Suppose the value 7 in the table were replaced by 70. Which leaf prediction would move more strongly: the mean or the median?
 4. Why might a robust loss such as the Huber loss be useful?
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Solution - A single leaf

Solution 1

For the targets

$$2, 3, 4, 7$$

the mean is

$$\bar{y} = \frac{2 + 3 + 4 + 7}{4} = 4$$

The residuals for $c = 4$ are

$$-2, -1, 0, 3$$

Therefore

$$\text{SSE} = (-2)^2 + (-1)^2 + 0^2 + 3^2 = 14$$

and

$$\text{MSE} = \frac{14}{4} = 3.5$$

Solution 2

For a general loss function

$$L(c) = \sum_i (y_i - c)^2$$

differentiating gives

$$\frac{dL}{dc} = -2 \sum_i (y_i - c)$$

At the minimum,

$$\sum_i (y_i - c) = 0$$

hence

$$Nc = \sum_i y_i$$

and therefore

$$c = \frac{1}{N} \sum_i y_i$$

So the mean is not an arbitrary convention: it is the constant prediction that minimizes squared error.

Solution 3

The MSE is measured in squared target units. RMSE has the same units as the target and is therefore often easier to interpret directly.

Squared error strongly penalizes large residuals because the residual is squared. For two different sets of residuals the same MSE must not be expected. An MSE of 0 would be the optimal outcome.

Solution 4

If absolute error is minimized instead, a median is an optimal constant prediction. MAE is therefore less sensitive to extreme outliers than MSE. Huber loss combines quadratic behavior for small residuals with approximately linear behavior for large residuals.

Part II - How does a regression tree find its first split?

We now construct a tree from data generated by a known function. Using a regression tree, we will then model the function of non-training data.

Consider

$$f(x) = x^2$$

on the interval

$$0 \leq x \leq 6$$

Exercise 2.1 - Generate the training data by hand

Observe the function at the integer positions

$$x = 0, 1, 2, 3, 4, 5, 6$$

and complete the following table:

x	$y = f(x) = x^2$
0	
1	
2	

x	$y = f(x) = x^2$
3	
4	
5	
6	

Plot these seven training points in a sufficiently large diagram and also sketch the true function $f(x) = x^2$.

Exercise 2.2 - Before splitting: the root leaf

Initially, the tree contains only one leaf. Therefore all seven observations receive the same prediction (i.e. the mean value as you have seen in Part I).

1. Compute the mean target value in the root leaf.
2. Draw this constant prediction in the same graph as $f(x)$.
3. Compute the SSE of the root leaf.

This is the error before the tree has made any split.

Exercise 2.3 - Which split points does the algorithm need to test?

The feature values are already sorted:

0, 1, 2, 3, 4, 5, 6

A split has the form

$$x < t$$

A threshold only changes the division of the training data when it moves from one side of an observed x value to the other.

Therefore it is sufficient to test thresholds t between neighboring observations. What are these in this case and why would testing, for example, both $t = 0.6$ and $t = 0.8$ be unnecessary for this dataset?

Exercise 2.4 - Follow one candidate split step by step

Take the candidate threshold

$$t = 2.5$$

The split rule is

$$x < 2.5$$

Step 1: Assign observations to the two child nodes

Write down the x and y values that go to the left child:

$x < 2.5$ y

Write down the x and y values that go to the right child:

$x \geq 2.5$ y

Step 2: Determine the prediction in each child

Compute

$$\hat{y}_L = \text{mean}(y_{\text{in left child}})$$

and

$$\hat{y}_R = \text{mean}(y_{\text{in right child}})$$

Step 3: Measure the error after the split

Compute

$$\text{SSE}_L = \sum_{i \in L} (y_i - \hat{y}_L)^2$$

$$\text{SSE}_R = \sum_{i \in R} (y_i - \hat{y}_R)^2$$

and finally

$$\text{SSE}_{\text{split}} = \text{SSE}_L + \text{SSE}_R$$

Compare this error with the SSE of the unsplit root leaf.

Did the split improve the model?

Exercise 2.5 - Let the tree choose the best split

The training algorithm repeats exactly the calculation above for every candidate threshold.

Complete the table.

threshold t	mean left $\hat{y}_L$	mean right $\hat{y}_R$	SSE_L	SSE_R	total SSE
0.5					
1.5					
2.5					

threshold t	mean left $\hat{y}_L$	mean right $\hat{y}_R$	SSE_L	SSE_R	total SSE
3.5					
4.5					
5.5					

The decision rule of the tree is:

”Choose the split with the smallest total child error”

Equivalently, choose the split with the largest error reduction

$$\Delta SSE = SSE_{\text{parent}} - (SSE_L + SSE_R)$$

1. Which threshold is selected?
2. Write down the two leaf predictions.
3. Draw the resulting depth-1 tree.

Using this graph, we are now able to draw a new prediction of the data into the graph of exercise 2.1. You should now see directly why the output of a regression tree is piecewise constant.

Exercise 2.6 - Split the left child

The first split is only the beginning. A regression tree applies the same procedure recursively inside the child nodes.

Take the left child produced by the best split.

1. List the observations contained in this child.
2. Determine all candidate thresholds that can still split these observations into two non-empty groups.
3. For every candidate, calculate the total child SSE.
4. Select the best split.

Exercise 2.7 - Split the right child

Repeat the same procedure for the right child.

If only two observations remain in this child, how many meaningful candidate thresholds are possible?

Determine the best split.

Exercise 2.8 - Draw the depth-2 tree

Draw the complete tree with two levels of splitting.

For every internal node, write

- the split condition,

and for every final leaf, write

- the constant prediction of that leaf.

Then draw the corresponding step function $\hat{f}(x)$ in the same coordinate system as

$$f(x) = x^2$$

Exercise 2.9 - Compare the approximations

Compare the following three models visually:

1. one root leaf with no split,
2. the tree after one split,
3. the depth-2 tree.

Discuss:

- Which model has the smallest training error?
 - Why does the approximation become finer as the tree grows?
 - Even though the exact function $f(x) = x^2$ is known and contains no noise, why does a finite regression tree generally not reproduce the complete function exactly?
 - What would happen if the tree were allowed to keep splitting until every training observation had its own leaf?
 - Would zero training error imply that the tree has learned the true function between the observed data points?
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Optional extension - Add measurement noise

Suppose the observations are no longer exactly $y = x^2$, but

$$y_i = x_i^2 + \epsilon_i.$$

You do not need to rebuild the complete tree.

Discuss qualitatively:

1. What happens if the tree is made very deep?
2. Could the tree begin to model the random fluctuations ϵ_i ?
3. Would a very small training error necessarily imply a small error on new data?

This is the basic connection between tree depth and overfitting.

Solution - The first split

Solution 1

The training data are

x	$y = x^2$
0	0
1	1
2	4
3	9
4	16
5	25
6	36

The root prediction is

$$\bar{y} = \frac{0 + 1 + 4 + 9 + 16 + 25 + 36}{7} = 13$$

Solution 2

The root SSE is

$$\text{SSE}_{\text{root}} = 1092$$

Solution 3

The relevant candidate thresholds are the midpoints between neighboring observations:

$$0.5, 1.5, 2.5, 3.5, 4.5, 5.5$$

Any two thresholds between the same neighboring observations produce exactly the same assignment of training points, which is why only one representative threshold is needed.

Solution 4

For $t = 2.5$:

Left child:

$x < 2.5$	y
0	0
1	1
2	4

$$\hat{y}_L = \frac{5}{3}.$$

Right child:

$x \geq 2.5$	y
3	9
4	16
5	25
6	36

with prediction

$$\hat{y}_R = 21.5.$$

The errors are approximately

$$\text{SSE}_L = 8.67,$$

$$\text{SSE}_R = 409,$$

and

$$\text{SSE}_{\text{split}} \approx 417.67.$$

This is much smaller than the unsplit error of 1092.

Solution 5

The full comparison is

threshold t	mean left	mean right	SSE_L	SSE_R	total SSE
0.5	0.00	15.17	0.00	894.83	894.83
1.5	0.50	18.00	0.50	654.00	654.50
2.5	1.67	21.50	8.67	409.00	417.67
3.5	3.50	25.67	49.00	200.67	249.67
4.5	6.00	30.50	174.00	60.50	234.50
5.5	9.17	36.00	474.83	0.00	474.83

Therefore the best root split is

$$x < 4.5.$$

The resulting depth-1 predictor is

$$\hat{f}(x) = \begin{cases} 6, & x < 4.5, \\ 30.5, & x \geq 4.5. \end{cases}$$

This is already a step function.

Solution 6

After the first split, the left child contains

$$x = (0, 1, 2, 3, 4) \quad y = (0, 1, 4, 9, 16)$$

Its candidate thresholds are

$$0.5, 1.5, 2.5, 3.5$$

The corresponding total SSE values are approximately

threshold	total SSE
0.5	129.00
1.5	73.17
2.5	33.17
3.5	49.00

Thus the left child selects

$$x < 2.5$$

Solution 7

The right child contains only

$$x = (5, 6) \quad y = (25, 36)$$

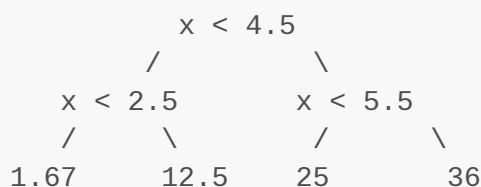
There is only one candidate threshold:

$$x < 5.5.$$

After that split each of these two observations occupies its own leaf, so the SSE of this branch becomes zero.

Solution 8

The depth-2 tree is therefore



The leaf predictions come from

$$\frac{0 + 1 + 4}{3} \approx 1.67,$$
$$\frac{9 + 16}{2} = 12.5,$$

and the single leaves 25 and 36.

The total training SSE is now approximately

$$33.17,$$

which is far below the depth-1 value

$$234.5.$$

Solution 9

Increasing depth therefore gives the tree more regions and a finer piecewise-constant approximation.

However, even a tree that perfectly fits all observed points does not automatically recover the exact function between those observations.

With noisy data, a very deep tree can begin to fit the noise rather than only the underlying relation. This is overfitting.

Part III - More than one input feature

A regression tree can choose not only where to split, but also which feature to split.

Consider the two-dimensional function

$$f(x_1, x_2) = x_1^2 + 4x_2.$$

We sample

$$x_1 \in \{0, 1, 2, 3\}, \quad x_2 \in \{0, 1\}.$$

Exercise 3.1 - Understand the function before building a tree

First fix $x_2 = 0$.

Then

$$f(x_1, 0) = x_1^2.$$

Now fix $x_2 = 1$.

Then

$$f(x_1, 1) = x_1^2 + 4.$$

1. Evaluate both functions at $x_1 = 0, 1, 2, 3$.
2. Complete the table.

x_1	x_2	$y = f(x_1, x_2)$
0	0	
1	0	
2	0	
3	0	
0	1	
1	1	
2	1	
3	1	

3. In separate coordinate systems, plot $f(x_1, 0)$ and $f(x_1, 1)$ as functions of x_1 .

Exercise 3.2 - Which feature is chosen first?

The tree may now test conditions involving either feature. For x_1 and x_2 , these are 4 thresholds in total, 3 for conditions involving x_1 and one for x_2 , thus, at the root node, the algorithm must compare four candidate splits.

Complete the following table by repeating the same steps as in Part II:

1. assign observations to the left and right child,
2. calculate the mean target in each child,
3. calculate the SSE in each child,
4. add the two errors.

feature	threshold	mean left	mean right	total SSE
x_1	0.5			
x_1	1.5			
x_1	2.5			
x_2	0.5			

Which feature and threshold are selected?

Draw this first split in the (x_1, x_2) feature plane.

- What does a split on x_1 look like geometrically?
 - What would a split on x_2 look like geometrically?
-

Exercise 3.3 - Grow one more level

Now allow each child node to split once more.

For each child:

1. determine which observations are present,
2. determine which candidate splits are still possible,
3. calculate their total SSE,
4. choose the best split.

Draw the resulting depth-2 tree.

Then again sketch the prediction as a function of x_1

- for $x_2 = 0$,
- for $x_2 = 1$.

How has the approximation changed?

What does this example tell you about how a regression tree combines information from several input features?

Solutions - Two input features

Solution 1

For

$$f(x_1, x_2) = x_1^2 + 4x_2,$$

the data are

x_1	x_2	y
0	0	0
1	0	1
2	0	4
3	0	9
0	1	4
1	1	5
2	1	8
3	1	13

For $x_2 = 0$,

$$f(x_1, 0) = x_1^2,$$

while for $x_2 = 1$,

$$f(x_1, 1) = x_1^2 + 4.$$

Thus the second curve is shifted upward by 4.

The root candidate thresholds are

$$x_1 < 0.5 \quad x_1 < 1.5 \quad x_1 < 2.5$$

and

$$x_2 < 0.5.$$

The split comparison is

feature	threshold	mean left	mean right	total SSE
x_1	0.5	2.00	6.67	97.33
x_1	1.5	2.50	8.50	58.00
x_1	2.5	3.67	11.00	49.33
x_2	0.5	3.50	7.50	98.00

Therefore the root selects

$$x_1 < 2.5$$

The depth-1 tree predicts the same value for $x_2 = 0$ and $x_2 = 1$ whenever the two observations fall into the same x_1 region, because the root split has not used x_2 yet.

For the left child, the best second split is

$$x_2 < 0.5$$

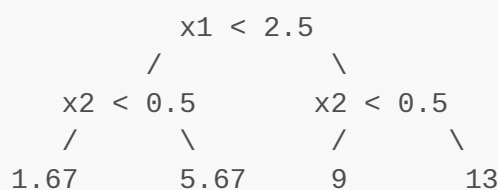
which separates the two shifted curves.

For the right child, the best split is also

$$x_2 < 0.5$$

which separates the values 9 and 13 exactly.

The resulting depth-2 tree can therefore use both input features:



The example illustrates an important property of ordinary decision trees:

”Each individual split uses only one feature, but different nodes can split on different features”

By combining several axis-aligned splits, the tree partitions a multidimensional feature space into rectangular regions.

Short conceptual questions

Question 1

What value does a regression-tree leaf predict when squared error is used?

Question 2

Why is that value the mean?

Question 3

What quantity is minimized when choosing a split?

Question 4

Where do the candidate thresholds come from for a continuous feature?

Question 5

Why is the prediction of a regression tree piecewise constant?

Question 6

What is the main effect of increasing tree depth?

Question 7

Why can a deep tree overfit noisy data?

Question 8

In a multidimensional input space, what additional choice must the tree make at every split?

Question 9

What shape do ordinary decision-tree regions have when only axis-aligned splits are used?

Solutions - Conceptual questions

1. With squared error, a leaf predicts the **mean target value** of the samples in that leaf.
2. The mean minimizes the sum of squared residuals.
3. A split is chosen to minimize the total error in its child nodes, or equivalently to maximize the reduction in error.

4. Candidate thresholds can be placed between neighboring sorted feature values.
5. Every leaf predicts a constant, so the overall prediction is piecewise constant.
6. Increasing depth creates more and smaller regions and usually reduces training error.
7. A deep tree can adapt to random fluctuations in the training sample.
8. With several input features, the tree must choose both the **feature** and the **threshold**.
9. With axis-aligned splits, the resulting regions are intervals in 1D and rectangles or hyperrectangles in higher dimensions.